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# **Computergestützte Inferenz und Analyse von PRDM9-unabhängigen meiotischen Rekombinations-Hotspots in Populationen der Europäischen Mönchsgrasmücke *Sylvia atricapilla***

Computational inference and analysis of prdm9 independent meiotic  
recombination hotspots in European Blackcap *Sylvia atricapilla* populations

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## **Masterarbeit**

im Ein Fach Masterstudiengang  
Fach Molekulare Biologie und Evolution  
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# 1 Introduction

## 2 Materials & Methods

### 2.1 Data Acquisition

To computationally identify and characterize prdm9-independent hotspots in European blackcap (*Sylvia atricapilla*), hereafter referred to as blackcap, on 15 March 2025 I retrieved:

- Population level blackcap whole genome variant call VCF, blackcap individual identifiers and inferred demography history data through SMC from (Ishigohoka and Liedvogel 2025).
- Blackcap genome assembly, GFF and genome sizes from NCBI with genome accession number GCF\_009819655.1 (Sayers et al. 2024).
- Zebra finch to blackcap lift-over chain file taeGut2ToGCF\_009819655.1.over.chain from the UCSC genome browser (Perez et al. 2024; Hinrichs et al. 2006).
- Zebra finch recombination rate maps from Singhal et al. 2015

As the demography data from Ishigohoka and Liedvogel 2025 was computed for each blackcap sub population, I merged the msmc2 table by taking the average of each input across all tables. I automated this process with a custom script `merge_msmc2_table.py`.

### 2.2 Population Genomics Statistics

To describe the different aspects of genomic variation, such as average pairwise sequence variation between individuals, frequency of polymorphic sites and deviation from natural evolution I computed key population genomics statistics: **Tajima's D**, **Window  $\pi$**  and **SNP density** using VCFTools (v0.1.16) (Danecek et al. 2011; Tajima 1989; Nei and Li 1979). To compute the statistics I first sampled 100 random individual identifiers using GNU `shuf` and saved them into a file called `n100_blackcap.panel`. Using this file, I then extracted the corresponding SNP records with a minor allele frequency maf of 0.05% and computed the statistics in window sizes of 1000, 5000, 10,000 and 100,000 base pairs (bp). Next, I plotted the genome-wide distribution of each statistic against their midpoints, which were calculated through the formula:  $\frac{Start+End}{2}$ . Lastly, divided the statistics into three equal quantile-based bins representing "low", "medium"

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and "high" values and visualized the proportion of each quantity per chromosome. I automated the visualization with a custom script called `plot_popgen_stats.py`.

## 2.3 Inference of Recombination Rate Maps

To compute the population scaled recombination rate maps in blackcaps, I utilized a demography-aware method to infer fine-scale recombination rates between SNPs called `pyrho` (Spence and Song 2019). I first precomputed the likelihood tables using `pyrho make_table` needed for estimating the population-specific recombination rate (Spence and Song 2019). The table was precomputed with the following parameters:

- Mutation rate  $\mu$  :  $4.6 \times 10^{-9}$ , as reported in Bascón-Cardozo et al. 2024
- Decimal relative tolerance: 0.1 as reported in Bascón-Cardozo et al. 2024
- Population size (i.e., number of haplotypes): 200
- Moran population size: 200
- The averaged MSMC2 demography history data.

I automated this process with a custom script, `make_table.sh`. Secondly, I optimized the `block_penalty` and `window_size`, two hyperparameters' required for the computation of the population scaled recombination rate, using `pyrho hyperparam` (Spence and Song 2019). I ran 10 simulations over the following grid of values:

- **Block Penalty:** 10, 20, 30, 40, 50
- **Window Size:** 25, 30, 50, 100, 1000

The optimization used the Averaged MSMC2 demography history and precomputed likelihood table from the previous step. This step was automated using `optimize_hyperparameters.sh`. `pyrho hyperparam` returns a table with the pearson and spearman correlation coefficients and the  $L2$  norm. I selected a **window size of 100** with a **block penalty of 20** as they maximized the pearson and spearman correlation coefficient while minimizing the  $L2$  norm, as specified by (Spence and Song 2019). I sampled the same 100 individuals from the VCF data as in section 2.2, but only with allele count of at least 2. Lastly, using the precomputed likelihood, sampled individual SNP data table, window size of 100 with a block penalty of 20 I computed population-scaled the recombination rate with `pyrho optimize` (Spence and Song 2019). This process was automated with a custom script called `pyrho_optimize.sh`

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## 2.4 Analysis of Recombination Rate Landscape Over The Blackcap Genome

To investigate the population scaled recombination rate landscape in blackcaps, I converted the raw recombination rate to cM/Mb by averaging over genomic windows of 10 kilobases and multiplying by  $10^8$ . Then, I calculated the midpoint of the genomic coordinates by the formula:  $\frac{Start+End}{2}$ . Lastly, I plotted the recombination rate in cM/Mb over the genomic midpoint coordinates.

section I investigated the distribution of recombination rate strength per chromosome. To do this, I sorted the recombination rate per chromosome by signal strength and divided it into equal tertiles ("low", "medium" and "high"). I visualized the data on a chromosomal basis using pie charts.

To gain a more indepth understanding of the recombination landscape I incorporated the gene density and gc content into the analysis. I extracted the genes from the blackcap gff file using a custom script `clean_raw_gff.py` and computed the gene density over genomic windows of size  $n$ . I generated genomic bins of size  $n$  using `PyRanges`, counted the number of genes overlapping with each window and divided by the window size and plotted the distribution of the gene density along the genome and binned the gene density into tertiles representing "low", "medium" and "high" to see the proportion To compute the gc content, I used `bedtools makewindows` to make genomic windows per each chromosome, `bedtools nuc` to compute the gc content over the windows, which I automated with the script `compute_gc_content.sh`. Next, I plotted the gc content over each chromosome and binned it into three tertiles ("low", "medium" and "high") values to visualize how the gc content is distributed over the genome. I automated the plotting in the steps above using the script `plot_popgen_stats.py`

To quantify the correlation between the population genomic statistics, gc content, gene density and the recombination rate I first intersected each feature with one another using `pyRanges`, computed the pearson correlation coefficient between each genomic feature using `SciPy` and visualized the results in a heatmap (Virtanen et al. 2020). I automated this analysis with a custom script called `compute_feature_correlation.py`.

Lastly, I investigated the average profile of the recombination rate over genomic features (UTR's, CDS, Exons, Intergenic Regions and Genes), which I extracted from the blackcap GFF using the

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`clean_raw_gff.py`. I used `deeptool`'s `computeMatrix` to calculate the average profile of the recombination rate over these regions and `plotHeatmap` to plot the average profile and heatmap. The average recombination rate profile was computed for  $\pm 20$  kb and zeros were not reported. The script for computing the matrix and plotting the profile are: `computeRhoMatrix.sh` and `plotHeatmap.sh`, respectively.

## 2.5 Inference of Meiotic Recombination Hotspots

To call recombination hotspots from recombination rate maps, I applied **Continuous Wavelet Transform (CWT)** to detect multi-scale peaks in the signal. To do so the wavelet is continuously rescaled across a grid of specified scales and shifted along the signal, producing coefficients that quantify local similarity between the signal and the wavelet shape at each position and scale. The higher the similarity, the higher coefficients (Singh, Rawat, and Raghuthaman 2022; González-Nuevo et al. 2006). Before computing the cwt with `PyWavelet`, I smoothed the recombination rate  $\rho$  signal using `SciPy`'s one dimensional gaussian filter (`gaussian_filter1d`) ( $\sigma = 5$ ) to remove excessive high frequency noise. Then, I generated 200 logarithmically spaced scales between 50 and 600. With an recombination rate genomic range of 22 bp this corresponds to genomic bin size between  $\sim 1.1$  kb to  $\sim 13.2$  kb. After computing the wavelet coefficients, I normalized the coefficients at each scale independently through z-score. Following this, I selected only those coefficients, which had a z-score:  $z > 2.0$  and persisted in at least 30% of the scales. The peaks were then identified using `SciPy`'s `find_peaks` with a minimum peak prominence of at least the 90<sup>th</sup> percentile. This process was automated using the script `find_hotspots.py`. Lastly, I used the package `cobind.py` to quantify the amount of overlap between the identified recombination hotspots and genomic features (UTRs', codons, genes, intergenic regions) (Ma et al. 2023)

To test if the wavelet transform methodology is also applicable to other `prdm9` independent recombining species I ran the same analysis on `LDHelmets` recombination maps generated by Singhal et al. 2015.

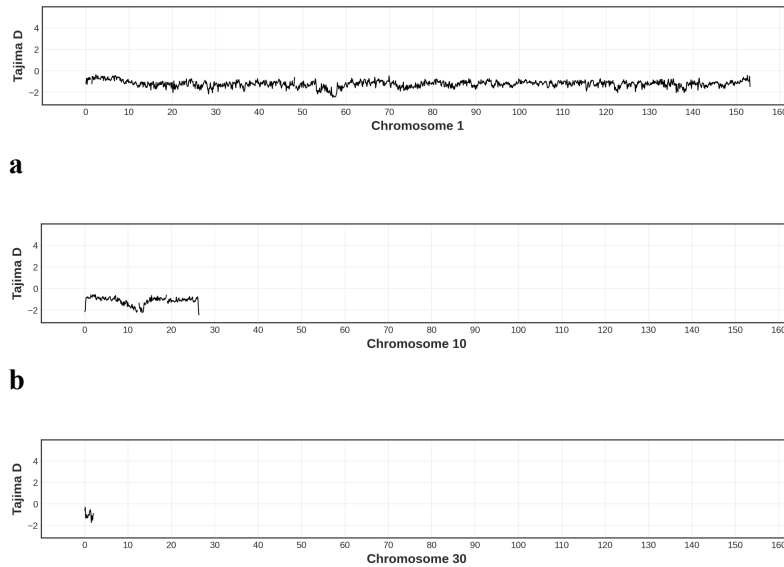
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## 3 Results

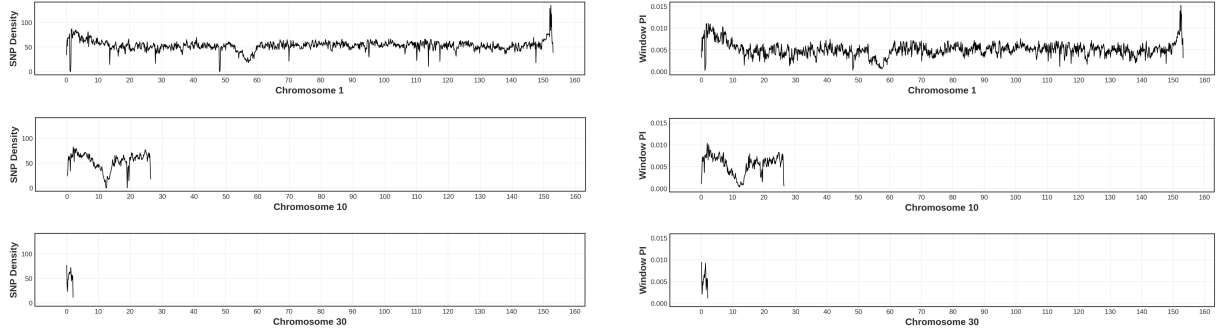
### 3.1 Population genomics statistics of the european blackcap metapopulation

To investigate the deviation from neutral evolution, average pairwise nucleotide diversity and snp density I computed three population genomics statistics: tajima  $D$ , diversity  $\pi$  and SNP density across the autosomal chromosomes on a blackcap metapopulation consisting of 100 individuals. Tajima's  $D$  showed two types of patterns across each of the 33 blackcap autosomal chromosomes (Figure 1, Supplementary Figure 1). The large chromosomes one through 12, showed relatively stable negative tajima  $D$  values across the chromosomal body, with the exception of one peak in chromosomes 2 and 4 at  $\sim 20Mb$  with a tajima  $D$  value of 0 and a sharp valley with a tajima  $D$  value of over -2 at  $\sim 7Mb$  on chromosome 6, and the end of chromosome 12 that contained several peaks of positive tajima  $D$  (Supplementary Figure 1). The chromosomal ends of these macrochromosomes had negative tajima  $D$  values which were close to 0.

The blackcap microchromosomes displayed opposite patterns, the subtelomeric regions had valleys beyond values of tajima  $D$  :  $-2$  while the chromosomal body showed stable yet tiny fluctuations in the tajima  $D$  values, which were negative yet closer to 0 (Supplementary Figure 1).



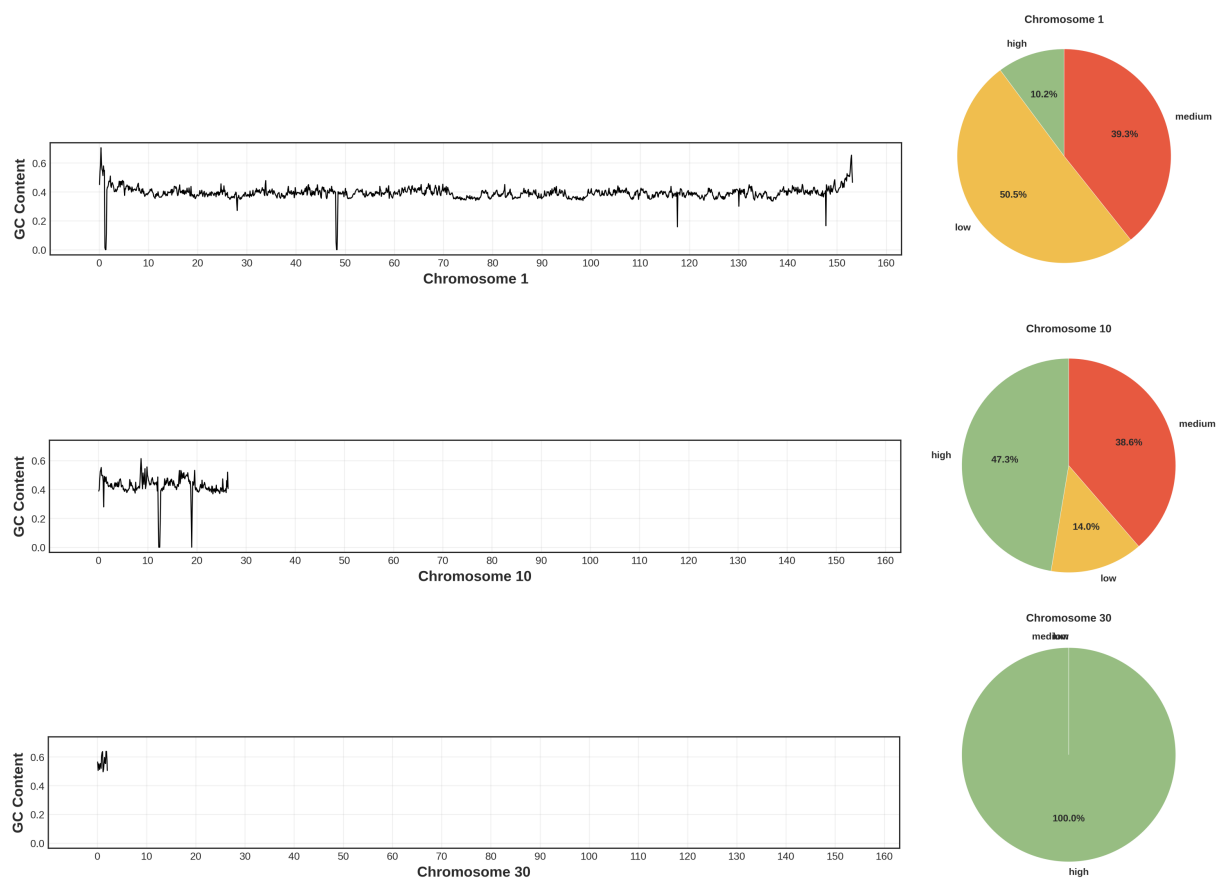
**Figure 1.** Tajima's  $D$  across chromosomes 1, 10, and 30 in 100 kb windows for a European Blackcap metapopulation (100 individuals from continental and island populations).



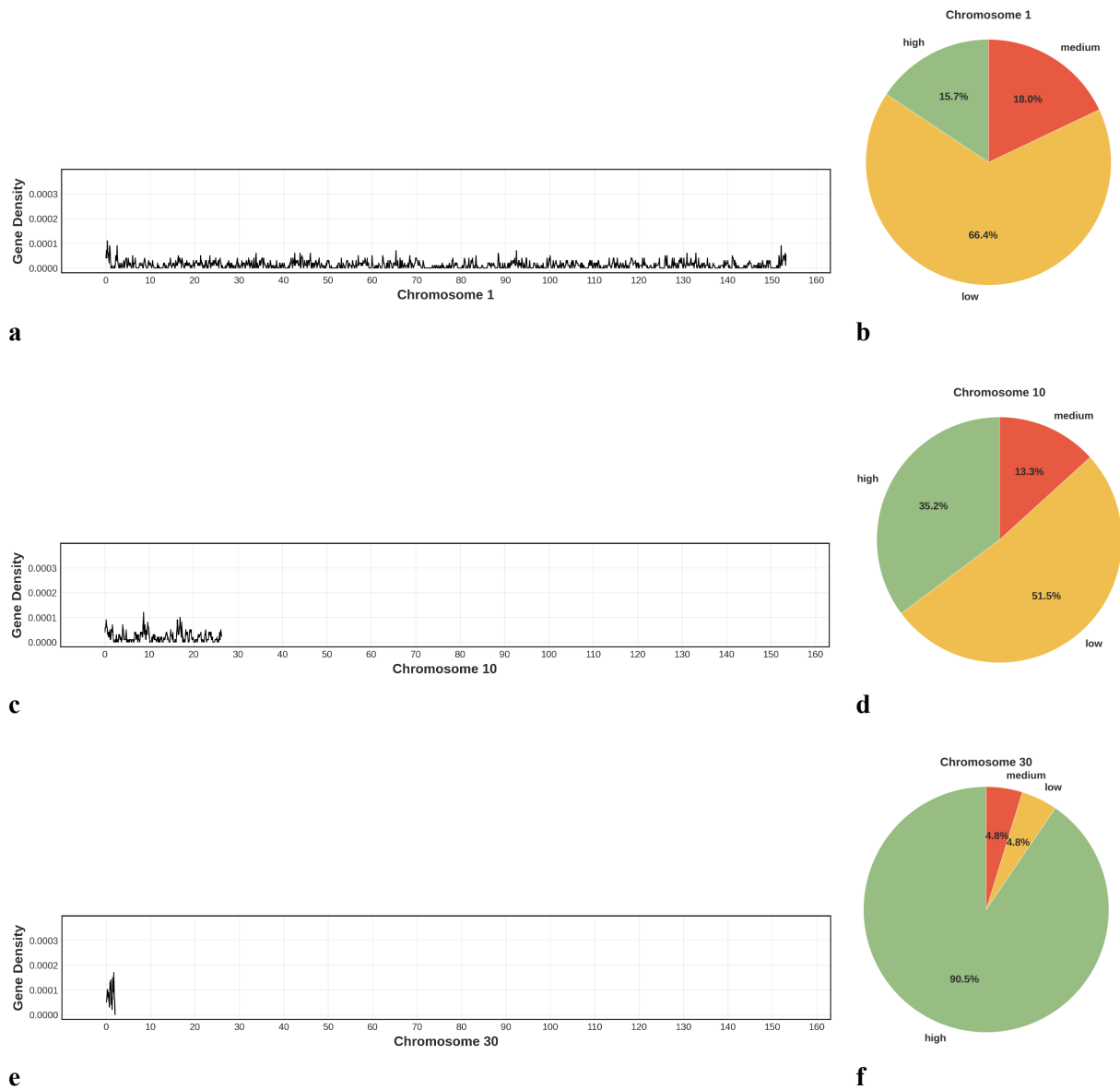
**Figure 2.** SNP density (left) and nucleotide diversity ( $\pi$ ) (right) across chromosomes 1, 10, and 30 in 100 kb windows.

Compared to tajima's  $D$ , the SNP density and nucleotide diversity  $\pi$  displayed more visible patterns across the chromosomes (Figure 2). The macrochromosomes had an average SNP density and nucleotide diversity  $\pi$  of 50 and 0.005 over 100 Mb window bins, respectively (Figure 2; Supplementary Figure 2 & 3). In addition to this, both values displayed regularly interspaced sharp valleys occurring at  $\sim 10$  Mb. Chromosome 1, in contrast to the other (Supplementary Figures 2 & 3).

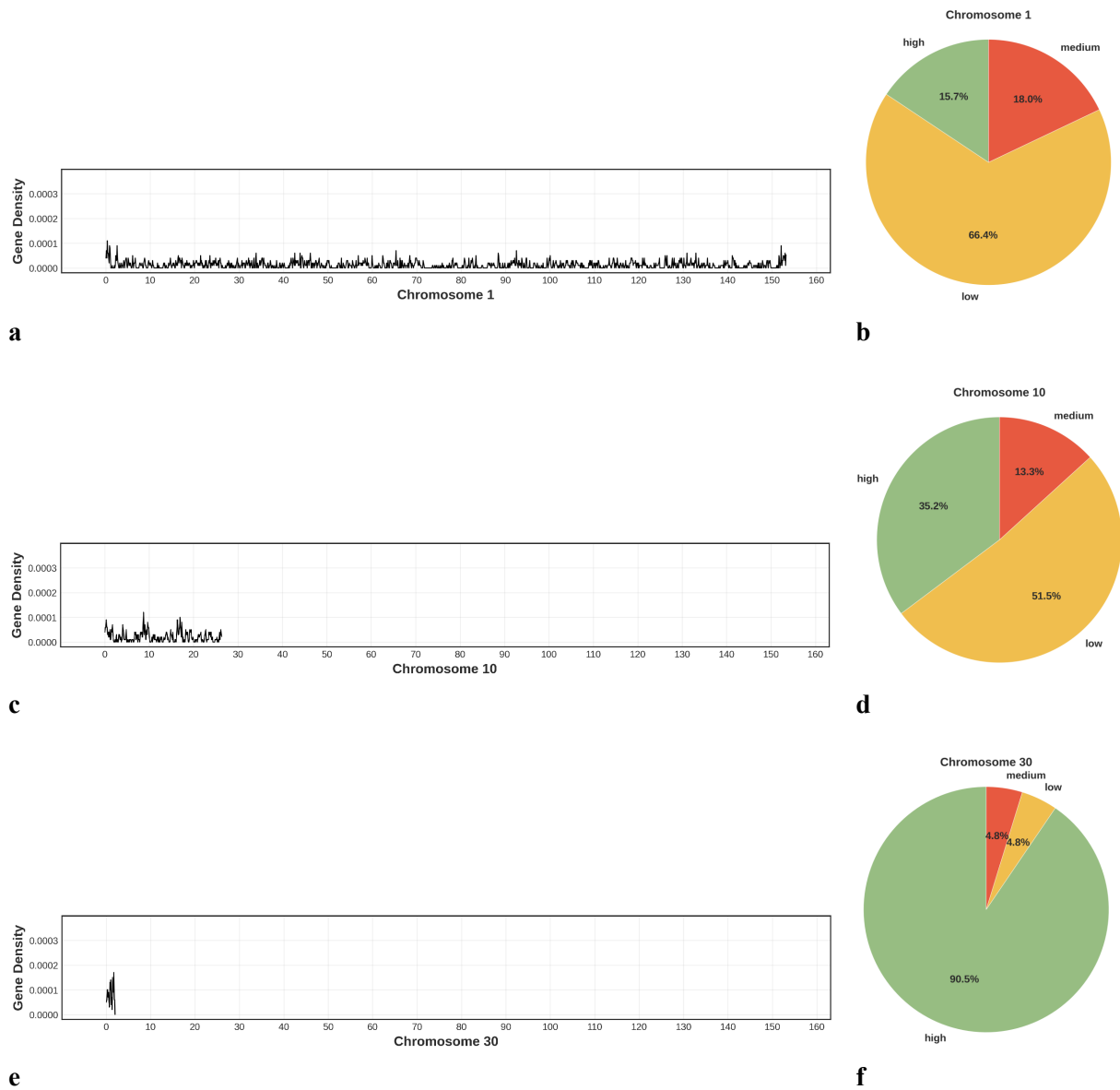




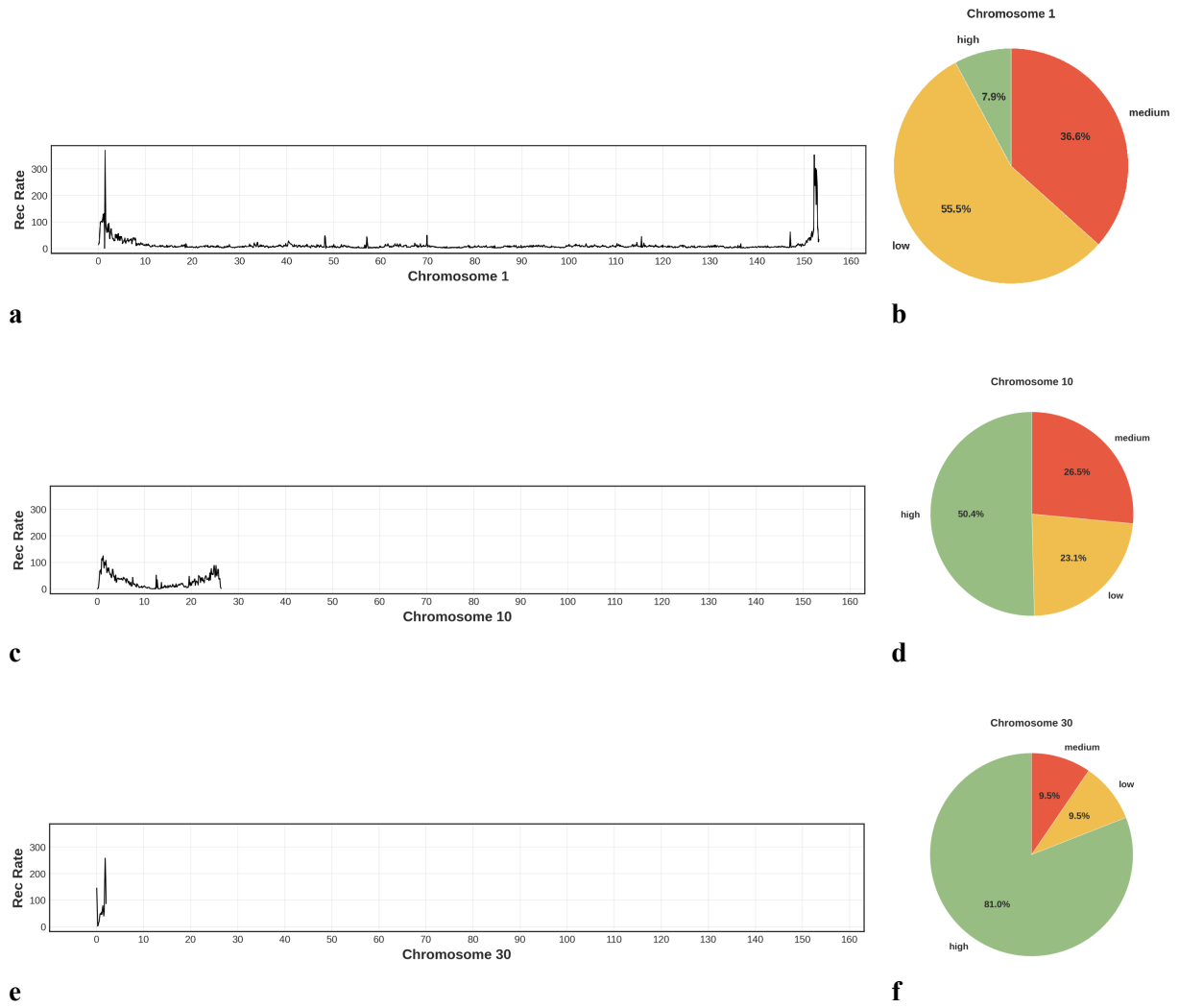
**Figure 3. (left)** GC content across chromosomes 1, 10, and 30 in 100 kb windows and **(right)** GC content binned into equal tertiles (*low, medium, high*).



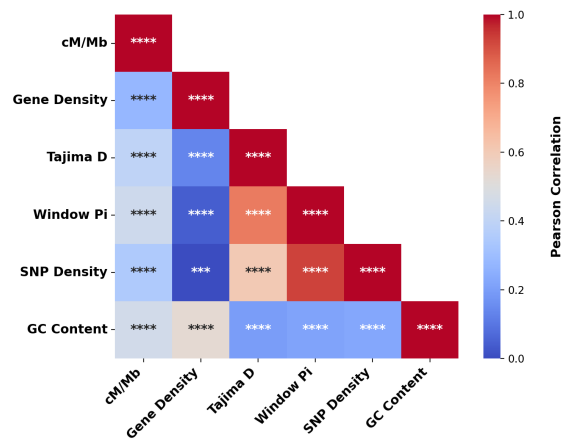
**Figure 4.** **a,c,e:** Gene density across chromosomes 1, 10, and 30 in 100 kb windows. **b,d,f:** Gene density binned into equal tertiles (*low*, *medium*, *high*).



**Figure 5.** **a,c,e:** Gene density across chromosomes 1, 10, and 30 in 100 kb windows. **b,d,f:** Gene density binned into equal tertiles (*low*, *medium*, *high*).



**Figure 6.** **a,c,e:** Recombination Rate ( $\frac{cM}{Mb}$ ) across chromosomes 1, 10, and 30 in 100 kb windows. **b,d,f:** Recombination Rate binned into equal tertiles (*low*, *medium*, *high*).



**Figure 7.** Heatmap showing the pearson correlation coefficient between several blackcap metapopulation features with corresponding  $p$ -values.

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## 4 Discussion

## 5 References

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## 6 Declaration

### Erklärung / Declaration

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbständig und ohne fremde Hilfe angefertigt und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe. Die eingereichte schriftliche Fassung der Arbeit entspricht der auf dem elektronischen Speichermedium.

*I hereby declare that I have prepared this thesis independently and without outside assistance. I have not used any sources or aids other than those indicated. The submitted written version of the thesis corresponds to the one on the electronic storage medium.*

Weiterhin versichere ich, dass diese Arbeit noch nicht als Abschlussarbeit an anderer Stelle vorgelegen hat.

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Datum, Unterschrift

*Date, Signature*